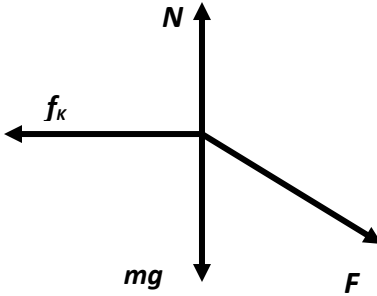


NO.	SOLUTION & ANSWER	MARKS															
1 (a)(i)	$988 \text{ km h}^{-1} = 988 \times \frac{10^3 \text{ m}}{3600 \text{ s}} = 274.44 \text{ m s}^{-1}$	GJU1															
1(a)(ii)	$60 \text{ } \mu\text{m} = 60 \times 10^{-6} \text{ m} = 6.0 \times 10^{-5} \text{ m}$	GJU1															
1(b)	<table border="1"> <thead> <tr> <th>Vector</th><th>x-component</th><th>y-component</th></tr> </thead> <tbody> <tr> <td>A</td><td> $A \sin 60^\circ$ $= 80 \sin 60^\circ$ $= 69.28 \text{ N}$ </td><td> $A \cos 60^\circ$ $= 80 \cos 60^\circ$ $= 40 \text{ N}$ </td></tr> <tr> <td>B</td><td> $B \cos 45^\circ$ $= 50 \cos 45^\circ$ $= 35.35 \text{ N}$ </td><td> $-B \sin 45^\circ$ $= -50 \sin 45^\circ$ $= -35.35 \text{ N}$ </td></tr> <tr> <td>C</td><td> $-C \cos 80^\circ$ $= -30 \cos 80^\circ$ $= -5.21 \text{ N}$ </td><td> $-C \sin 80^\circ$ $= -30 \sin 80^\circ$ $= -29.54 \text{ N}$ </td></tr> <tr> <td>Magnitude, F_R</td><td>$\sum F_x = 99.42 \text{ N}$</td><td>$\sum F_y = -24.89 \text{ N}$</td></tr> </tbody> </table> <i>Magnitude, $F_R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$</i> $= \sqrt{(99.42)^2 + (-24.89)^2}$ $= 102.49 \text{ N}$ Direction , θ $\tan \theta = \frac{\sum F_y}{\sum F_x}$ $= \frac{-24.89}{99.42}$ $= 0.25$ $\theta = 14.05^\circ$ below positive x-axis	Vector	x-component	y-component	A	$A \sin 60^\circ$ $= 80 \sin 60^\circ$ $= 69.28 \text{ N}$	$A \cos 60^\circ$ $= 80 \cos 60^\circ$ $= 40 \text{ N}$	B	$B \cos 45^\circ$ $= 50 \cos 45^\circ$ $= 35.35 \text{ N}$	$-B \sin 45^\circ$ $= -50 \sin 45^\circ$ $= -35.35 \text{ N}$	C	$-C \cos 80^\circ$ $= -30 \cos 80^\circ$ $= -5.21 \text{ N}$	$-C \sin 80^\circ$ $= -30 \sin 80^\circ$ $= -29.54 \text{ N}$	Magnitude, F_R	$\sum F_x = 99.42 \text{ N}$	$\sum F_y = -24.89 \text{ N}$	K1 K1 K1 GJU1 G1 JU1
Vector	x-component	y-component															
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	JUMLAH	8															

NO.	SOLUTION & ANSWER	MARKS
2(a)(i)	$v_{0-2} = \frac{18-0}{2-0} = 9 \text{ m s}^{-1}$ $v_{2-7} = \frac{18-18}{7-2} = 0 \text{ m s}^{-1}$ $v_{7-10} = \frac{0-18}{10-7} = -6 \text{ m s}^{-1}$	GJU1 GJU1 GJU1
2(a)(ii)		Shape graph D2 Axis unit D1 Label D1
2(b)	$s = ut + \frac{1}{2}at^2$ $30 = (u)(3) + \frac{1}{2}(2)(3)^2$ $u = 7 \text{ m s}^{-1}$	G1 JU1
2(c)	$s = ut + \frac{1}{2}at^2$ $= (15)(2) + \frac{1}{2}(-9.81)(2)^2$ $= 10.38 \text{ m}$ above released point	G1 JU1 J1
	JUMLAH	12

NO.	SOLUTION & ANSWER	MARKS
3(a)(i)	$J = \text{Area under graph}$ $= (12)(15 + 10)(30)$ $= 37.5 \text{ Ns}$	GJU1
3(a)(ii)	$J = mv - mu$ $37.5 = 5v - 5(0)$ $v = 7.5 \text{ ms}^{-1}$	GJU1
3(b)(i)	$\Sigma p_{ix} = \Sigma p_{fx}$ $m_1 u_{1x} + m_2 u_{2x} = m_1 v_{1x} + m_2 v_{2x}$ $(2000)(80) + 0 = (2000)(4.33) + 860 v_{2x}$ $v_{2x} = 175.98 \text{ ms}^{-1}$	K 1 G 1 JU 1
3(b)(ii)	$\Sigma p_{iy} = \Sigma p_{fy}$ $m_1 u_{1y} + m_2 u_{2y} = m_1 v_{1y} + m_2 v_{2y}$ $0 = (2000)(5)\sin 30 + 860 v_{2y}$ $v_{2y} = -5.814 \text{ ms}^{-1}$	G 1 JU 1
3(b)(iii)	$v = \sqrt{v_{2x}^2 + v_{2y}^2}$ $= \sqrt{(175.98)^2 + (-5.814)^2}$ $= 176.08 \text{ ms}^{-1}$	K 1 GJU 1
	JUMLAH	9

NO.	SOLUTION & ANSWER	MARKS
4(a)	$\Sigma F_y = 0$ $T_p \sin 60^\circ - 18(9.81) = 0$ $T_p = 203.9 \text{ N}$ $\Sigma F_x = 0$ $T_Q - 203.9 \cos 60^\circ = 0$ $T_Q = 101.95 \text{ N}$	K1 G1 JU1 GJU1
4(b)(i)	 <i>Label and direction</i> All correct ----- 3 marks 3 correct ----- 2 marks 2 correct----- 1 marks	 D3
4(b)(ii)	$\Sigma F_x = ma$ $43 \sin 65^\circ - f_k = 2.5(0.6)$ $f_k = 37.47 \text{ N}$	K1 G1 JU1
4(b)(iii)	$\Sigma F_y = 0$ $N - 2.5(9.81) - 43 \cos 65^\circ = 0,$ $N = 42.7 \text{ N}$ $f_k = \mu N$ $\mu = \frac{f_k}{N} = \frac{37.47}{42.7} = 0.88$	 G1 GJU1
	JUMLAH	12

NO.	SOLUTION & ANSWER	MARKS
5(a)(i)	$T = \frac{\text{time taken}}{\text{no. of revolution}} = \frac{155}{95}$ $= 1.63 \text{ s}$	GJU1
5(a)(ii)	$\frac{\theta}{t} = \frac{2\pi}{T}$ $\frac{\theta}{5(60)} = \frac{2\pi}{1.63}$ $\theta = 1155.3 \text{ rad OR } 183.87 \text{ rev}$	K1 GJU1
5(b)(i)	$20 \text{ rpm} = \frac{20(2\pi)}{60}$ $= 2.09 \text{ rad s}^{-1} \text{ OR } \frac{2\pi}{3} \text{ rad s}^{-1}$	GJU1
5(b)(ii)	$F_c = f_k \rightarrow \mu_k mg = mr\omega^2$ $(0.45)(9.81) = r\left(\frac{2\pi}{3}\right)^2$ $r = 1.01 \text{ m}$	K1 GJU1
5(c)(i)	$r = l \sin \theta$ $= 0.68 \sin 35^\circ$ $= 0.39 \text{ m}$	GJU1
5(c)(ii)	$\tan \theta = \frac{F_c}{mg} = \frac{mv^2}{mrg}$ $\tan 35^\circ = \frac{v^2}{0.39(9.81)}$ $v = 1.64 \text{ m s}^{-1}$	K1 GJU1
	JUMLAH	9

NO.	SOLUTION & ANSWER	MARKS
6(a)(i)	$\omega_0 = 0.150 \text{ revs}^{-1} \times 2\pi = 0.300\pi \text{ rads}^{-1}$	G1
	$\alpha = 0.750 \text{ revs}^{-2} \times 2\pi = 1.50\pi \text{ rads}^{-2}$	G1
	$\omega = \omega_0 + \alpha t$	
	$= 0.300\pi + 1.50\pi(4.00)$ $= 19.8 \text{ rads}^{-1}$	JU1
6(a)(ii)	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$	
	$= (0.300\pi)(4) + \frac{1}{2}(1.50\pi)(4)^2$	
	$= 13.2\pi$	G1
	$\text{Number of revolutions} = \frac{\theta}{2\pi} = \frac{13.2\pi}{2\pi} = 6.6 \text{ revolutions}$	J1
6(a)(iii)	$v = r\omega$ $= (0.400)(19.8)$ $= 7.92 \text{ ms}^{-1}$	GJU1
6(b)	$\omega = \omega_0 + \alpha t$	
	$0 = 0.300\pi + (-1.50\pi)t$	G1
	$t = 0.2 \text{ sec}$	JU1
6(c)	Considering pivoted point is located at the fulcrum fulcrum,	
	$\tau_{\text{nett}} = 0$	K1
	$\tau_{T_1} + \tau_{T_2} = 0$	
	$T_1 r_1 \sin 90 - T_2 r_2 \sin 90 = 0$	G1
	$(0.170)(9.81)r_1 - (0.200)(9.81)(0.20) = 0$	G1
	$r_1 = 0.235 \text{ m}$	JU1
	JUMLAH	12

NO.	SOLUTION & ANSWER	MARKS
7(a)(i)	$\frac{dT}{dx} = \frac{323.15 - 423.15}{0.4}$ $\frac{dT}{dx} = -250 K m^{-1}$	G1 JU1
7(a)(ii)	$\frac{dQ}{dt} = -kA \frac{dT}{dx}$ $\frac{dQ}{dt} = -(210)(4 \times 10^{-4})(-250)$ $\frac{dQ}{dt} = 21 W$	G1 JU1
7(b)(i)	$v_{rms} = \sqrt{\frac{3kT}{m}}$ $600 = \sqrt{\frac{3(1.38 \times 10^{-23})(298.15)}{m}}$ $m = 3.43 \times 10^{-26} kg$	G1 JU1
7(b)(ii)	$v_{rms} = \sqrt{\frac{3kT}{m}}$ $750 = \sqrt{\frac{3(1.38 \times 10^{-23})T}{3.43 \times 10^{-26}}}$ $T = 466.03 K$	G1 JU1
7(c)(i)	adiabatic process $Q = 0$ $Q = \Delta U + W$ $0 = \Delta U + 5000$ $\Delta U = -5000 J$	K1 G1 JU1
7(c)(ii)	$Q = \Delta U + W$ $0 = 3000 + W$ $W = -300 J$	G1 JU1

7(d)(i)	$Q = \Delta U + W$ $10000 = \Delta U + 6000$ $\Delta U = 4000J$	G1 JU1
7(d)(ii)	$W = 0$ $Q = \Delta U + W$ $-2000 = \Delta U + 0$ $\Delta U = -2000J$	K1 G1 JU1
	JUMLAH	18